Flow Field and Combustion Characterization of Premixed Gas Turbine Flames by Planar Laser Techniques

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Lean premixed natural gas/air flames produced by an industrial gas turbine burner were analyzed using laser diagnostic methods. For this purpose, the burner was equipped with an optical combustion chamber and operated with preheated air at various thermal powers P, equivalence ratios Φ, and pressures up to $p=6$ bars. For the visualization of the flame emissions OH^* chemiluminescence imaging was applied. Absolute flow velocities were measured using particle image velocimetry (PIV), and the reaction zones as well as regions of burnt gas were characterized by planar laser-induced fluorescence (PLIF) of OH. Using these techniques, the combustion behavior was characterized in detail. The mean flow field could be divided into different regimes: the inflow, a central and an outer recirculation zone, and the outgoing exhaust flow. Single-shot PIV images demonstrated that the instantaneous flow field was composed of small and medium sized vortices, mainly located along the shear layers. The chemiluminescence images reflected the regions of heat release. From the PLIF images it was seen that the primary reactions are located in the shear layers between the inflow and the recirculation zones and that the appearance of the reaction zones changed with flame parameters.
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Keywords: lean premixed combustion, swirl flow, gas turbine combustor, high pressure, NO_x reduction, OH^* chemiluminescence, laser diagnostics, OH PLIF, PIV, reaction zone, heat release, turbulent flow field, vortex core, validation data

1 Introduction

The concept of lean premixed combustion is nowadays most often applied in stationary gas turbines (GTs) in order to achieve very low NO_x levels. However, the full potential of this concept for NO_x reduction can often not be exploited because effects of unmixedness, coherent flow structures, or combustion instabilities limit its applicability. In order to gain a better understanding of the complex interactions between the turbulent flow field and chemical reactions in GT-flames, detailed measurements under GT-relevant conditions are necessary. Here, the best available measurement techniques are nonintrusive laser-based methods. The results of such measurements play also an important role for the verification of numerical simulations. Especially in the field of turbulent combustion under high-pressure conditions there is a need of numerical tools, which are capable of correctly determining the flame behavior by taking into account phenomena such as the local extinction and re-ignition. To be able to validate existing models and to develop new numerical approaches, diverse comprehensive sets of measurement data are required. The ideal validation database would contain information about all basic flame properties such as the distributions of species concentrations, temperature, and flow velocity. Furthermore, the simulation of instationary combustion needs highly time-resolved measurements or discrete single-shot data, allowing statements about statistical and coherent deviations from the average flame behavior.

Laser-based measurements in high-pressure GT model combustors are expensive and difficult to perform [1,2]. Due to the limited optical access and the high thermal loads, robust laser techniques are preferably applied. For temperature measurements, coherent anti-Stokes Raman scattering (CARS) [3] and a two-line OH laser-induced fluorescence (LIF) [4] have been used. For the visualization of hot and reacting gases, the planar LIF of OH was frequently applied [5–9]; the LIF was also used for the detection of NO [10,11]. Flow field measurements were mostly performed by laser Doppler velocimetry (LDV) [12–14]. The addition of seeding particles to the flow and their contamination of the windows present an additional challenge in high-pressure GT model combustors. This is even more serious for the application of particle image velocimetry (PIV) and is a main reason for the relatively few publications describing PIV measurements at high-pressure GT conditions [15–17]. However, 2D images of the instantaneous flow field enable new insights into the dynamics of turbulent swirl flames, which are important for the understanding of combustion instabilities.

In the experiments described in this paper, an industrial GT burner was equipped with an optical combustion chamber and was installed in the high-pressure test rig of the German Aerospace Center in Stuttgart. The following diagnostic methods were applied in this project: Chemiluminescence imaging provided information about the distribution of electronically excited OH radicals (OH^*), which is a marker for the reaction zones. OH radicals in the electronic ground state were mapped using PLIF, and the flow field was measured by PIV. In addition to the optical techniques, CO, CO_2, NO, NO_2, and SO_2 were determined by probe techniques in the exhaust, and pressure variations in the combustion chamber were registered by dynamic pressure sensors. By these measurements, the behavior of lean premixed flames was characterized in detail for different pressures, thermal powers, and equivalence ratios. This paper focuses on the implementation and the results of the optical measurements. Parameter studies are pre-

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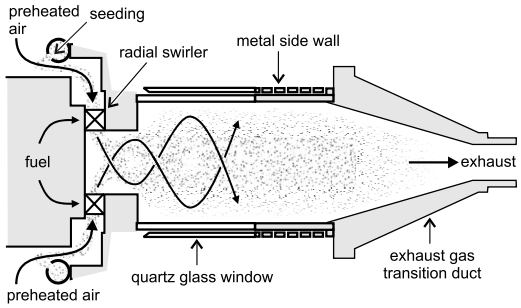

Fig. 1 Cut-away drawing of the combustor. The preheated air is seeded with TiO_2 particles for PIV measurements. A swirl burner mixes the air with natural gas. Each window consists of two quartz plates with a cooling air flowing in between.

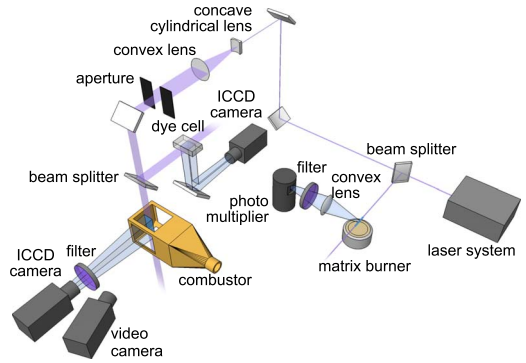

Fig. 2 Experimental setup for the imaging of the OH PLIF and the OH* chemiluminescence. One ICCD camera images the fluorescence, while the other one records the laser intensity profile.

sented in extracts, whereas probe measurements are not published. One main goal of this complete study was a better understanding of this combustor and the flame characteristics, e.g., the region of flame stabilization, the turbulent structures, the mixing characteristics, and pollutant emissions.

The work presented here is part of an ongoing research project, which will be continued by laser Raman measurements of the joint probability density functions (PDFs) of the temperature and major species concentrations. The final goal is the establishment of an experimental database, which allows also the identification and quantification of effects of unmixedness and finite-rate chemistry. Such "complete" data sets are currently available only for flames at atmospheric pressure, e.g., for turbulent nonpremixed flames [18] or (partially) premixed swirling flames [19,20].

2 Combustor and Test Rig

One aim of this project is to perform the experiments under conditions as close to the industrial case as possible. To meet this objective, an unscaled industrial swirl burner [21] was equipped with a specially designed optical combustion chamber (Fig. 1). It is capable of operation at a thermal power of 1 MW and provides extensive optical access for laser diagnostics. The inner dimensions of the square combustor are based on the size of a commercial gas turbine combustor. Inside, natural gas with preheated air is burnt in a lean premixed combustion. The air preheating temperature was 400°C for all operating points. To reduce the thermal load of the quartz glass, double-glazed windows with cooling air flowing between the glass plates have been installed. The rest of the combustion chamber sidewalls are cooled by impingement cooling. A water-cooled transition duct guides the exhaust gas into the rig's exhaust channel. For the flow field measurements, the gas in the combustion chamber had to be seeded with particles with a diameter of 1 μm. Therefore, a perforated ring tube was installed in the plenum upstream of the radial swirler. During the particle imaging it was used to add a constant flow of seeded air (at a mass flow of up to 4 g/s) to the preheated air. All measurements were performed at pressures p between 1.5 bars and 6 bars and burner pressure drops Δp between 1% and 3%, where the percentage refers to the absolute pressure upstream of the burner.

This combustor was installed at the high-pressure test rig facility HBK-S of the Institute for Combustion Technology in Stuttgart. The pressure vessel of the rig was equipped with glass windows for optical access and several probes for the measurement of temperature, absolute pressure, differential pressure, and acoustics. The concentrations of several species in the exhaust gas were monitored and recorded during every operation.

3 Experimental Implementation

All activities at the test rig facility are organized in measurement phases of several weeks. In the first phase of the project, the imaging of chemiluminescence and PLIF was accomplished, whereas the flow field studies took place during a second phase.

3.1 OH* Chemiluminescence and OH PLIF Imaging Setup. Figure 2 shows a schematic of the diagnostic system, which was used in the first measurement phase. The laser system consisted of a dye laser pumped by a Nd:YAG laser. The output of the dye laser was frequency doubled and tuned to a wavelength of 283.55 nm, which is resonant with the $Q_1(8)$ line of the A-X (1–0) band of the OH radical. In order to facilitate the adjustment of the laser wavelength and to control its stability, a small portion of the beam was coupled out by a beam splitter and irradiated into a flat laminar methane-air flame stabilized on a matrix burner. The OH LIF signal from this flame was detected by a photomultiplier tube equipped with a spectral bandpass filter that is transparent for wavelengths between 300 nm and 325 nm. The majority of the laser light passes the beam splitter and reaches two lenses, which are set up with a distance of about 64 mm from each other. The first lens ($f=-50$ mm) is shaped cylindrically, whereas the second one ($f=750$ mm) is a spherical lens. This optics produces a laser light sheet, which is guided vertically through the center of the combustion chamber. The focus of the sheet is located in the symmetry axis of the chamber. The LIF of OH radicals is imaged with an intensified charge-coupled device (ICCD) camera equipped with a bandpass filter identical to the one in front of the photomultiplier tube. To correct the PLIF images for the intensity distribution within the laser light sheet, a beam splitter redirects a small portion of the laser sheet into a dye cell. The fluorescence from this cell was recorded by a second ICCD and used for the correction. As the width of the laser sheet is about two or three times smaller than the size of the field of view, it was shifted in the axial direction in order to perform measurements at two different sheet positions. This is done by simultaneously moving all optical elements and detectors except the first beam splitter and the matrix burner with the photomultiplier. The processing of the raw images includes background correction, camera sensitivity correction, and laser intensity profile correction. After these processes the two OH concentration maps from the two corresponding measurements with different sheet positions were merged into one mapping of the whole region of interest.

The imaging of the OH* chemiluminescence was performed with the same detection system as used for PLIF. Details of the measuring technique are listed in Table 1.

3.2 PIV Setup. The PIV method calculates the vectors of the local streaming velocity by analyzing the displacement of very small seeding particles during a short time span. For measuring this displacement, the particles are illuminated with a double-pulsed laser sheet. A schematic of the PIV setup is shown in Fig. 3. The flashlamp-pumped Nd:YAG laser used in this setup emits

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Table 1 Technical details of the PLIF and chemiluminescence diagnostics

OH* chemiluminescence measurement	
Exposure time	40 μm
Frame rate	3 Hz
Effective image size	512×446 pixels
Bandpass filter	300–325 nm
OH PLIF measurement	
Exposure time	400 ns
Laser pulse duration	8 ns
Laser pulse energy	4.6–5.5 mJ
Laser wavelength	283.55 nm
Effective image size	359×446 pixels
Bandpass filter	300–325 nm
ICCD camera specifications (both cameras)	
Dynamic range	16 bits
CCD chip size	512×512 pixels
Objective (flame)	UV Halle, 100 mm, $f/2$
Objective (dye cell)	UV Nikkor, 105 mm, $f/4.5$

short double-pulses at $\lambda=532$ nm that are triggered synchronously with the exposure gates of the PIV camera (Table 2). A combination of two cylindrical lenses ($f_1=-12.7$ mm, $f_2=200$ mm) converts the shape of these pulses into a laser sheet wide enough to cover the whole field of view. 418 mm behind the cylindrical lenses a spherical lens ($f=1000$ mm) focuses the sheet and reduces its planar divergence. Similar to the PLIF setup, a large bending mirror guides the laser light vertically through the central plane of the combustor. The objective of the PIV camera is covered with a narrow band filter (532 nm), reducing the influence of disturbing light sources. The accuracy of the measured velocities depends on the seeding particle density, the degree of window contamination, and the turbulence of the flow. On the opposite side of the test rig an ICCD camera with an adequate filter for OH* chemiluminescence is mounted, providing images that can be compared with the results of the previous measurement phase, in order to check whether the flame behavior is identical.

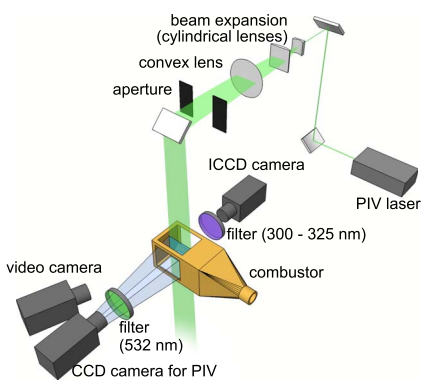

Fig. 3 Experimental setup for the PIV. An additional ICCD camera observes the OH* chemiluminescence. The video camera is used for control purposes only.

Table 2 Technical details of the PIV setup

PIV measurement	
Laser pulse duration	5 ns
Laser pulse energy	120 mJ
Laser sheet width	1 mm
Double-pulse delay	10–15 μs
PIV repetition rate	5 Hz
Seeding material	TiO_2(1 μm)
PIV camera objective	Nikkor, 50 mm, $f/5.6$

At good PIV measurement conditions, an interrogation window size of 12×12 pixels could be achieved. The accuracy of the single-shot velocity estimates can reach ±0.7 m/s (assuming a subpixel accuracy of particle displacement of 0.1 pixel and neglecting the influence of unrecognized erroneous vectors).

3.3 Experimental Challenges. As temperatures in the combustion chamber increase above 1600°C, the windows of the chamber have to sustain very high heat loads. Although the cooling system reduces the temperature of the glass, the surface of the interior window is still heated up to 1000°C. At this temperature the glass begins to soften and its surface becomes rough and increasingly opaque. Impurities on the window surface glow red and orange. This thermal radiation does not affect the measured chemiluminescence, which is in the UV regime and can be isolated using a spectral filter. More interfering is the effect the high temperature has on the opacity of the glass. The temperature distribution over the glass window is asymmetric as the swirled flow rotates through a noncylindrical combustion chamber. This in turn leads to an asymmetry in the opacity of the windows as the aforementioned roughness increases. The majority of the asymmetry in the chemiluminescence data can be ascribed to this asymmetric window transparency. Additionally, single particles on the window surface are reproduced as small spots in the chemiluminescence images.

During the PIV measurements the window staining was significantly increased by seeding particles adhering to the quartz glass. As a consequence, the visibility of the particles flowing through the central plane of the combustor decreased, while impurities accumulated on the window, producing more and more stationary spots in the image. Although stationary background signals in the series of single shots could be mostly eliminated by applying special filter routines, the described process caused a rapid degradation of the image quality. Below a certain signal-to-noise ratio the cross correlations calculated by the analyzing software were too falsified to enable the detection of the correct flow vector, resulting in an erroneous vector field. Depending on the window opacity this caused the single-shot analysis either to produce single blank regions or to completely fail. The former case might still be usable for calculating the temporal average of the flow field. However, if an averaged flow field is composed out of many low-quality single shots, it can contain artificially lowered absolute values of the velocity due to erroneous vectors that statistically do not favor any direction. This problem often can be bypassed by applying a method called "ensemble correlation" or "sum of correlation" (SOC). This method first sums up the cross correlations of all single shots and calculates the vector field afterward [22]. The result is the flow field with the maximum likelihood, which for the very most cases is practically identical with the temporal average of the flow field. The section of this paper about the analysis of the temporally averaged data does not differentiate between the temporal average and the results produced by the use of the SOC.

4 Experimental Results

The following four remarks should be kept in mind while interpreting the experimental results.

- Chemiluminescence imaging is a line-of-sight technique. This means that the detected signal intensity is not emitted from a certain excitation plane (as with PLIF or PIV). The emission is rather integrated along the whole line-of-sight. However, for the central parts of the flame we can assume a cylinder symmetry of the averaged OH* distribution, which allows us to apply the

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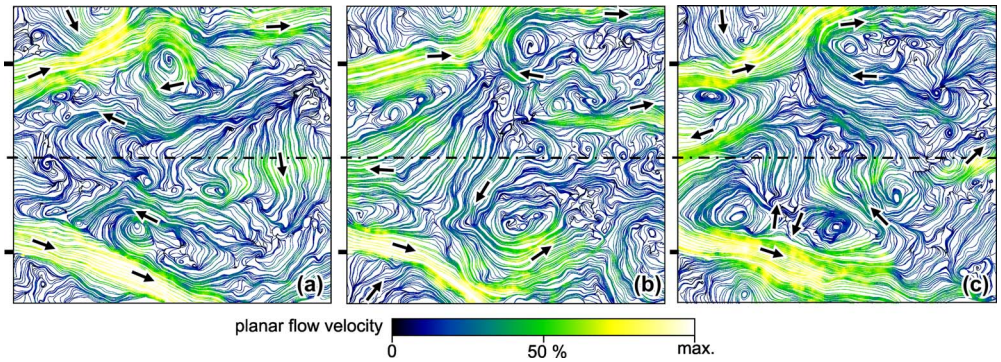

Fig. 4 Streamline plots of three successive instantaneous PIV measurements at $p=3$ bars, $\Delta p=1\%$, and $\Phi=0.59$. The size of the burner nozzle is indicated by two marks on the left side of each image. The PIV repetition rate was 5 Hz. Arrows indicate the local flow direction.

Abel inversion. This algorithm deconvolutes the image data in order to calculate the distribution of the signal intensity on a radial section plane.

•The color scale of all figures showing chemiluminescence or PLIF in this paper is scaled from zero intensity (black) to the maximum intensity of the particular image (white).

•The averaged data in this article that are described as temporally averaged are, in fact, ensemble averaged over a set of single-shot measurements that were performed in succession under constant operating conditions. Since each averaged image is composed out of 200 single shots, which were recorded at a frame rate of about 3–5 Hz, we can assume the ensemble-average to be identical with the temporal average. For clarity reasons the latter expression will be used here.

•All images are background corrected by subtracting an image with background intensity. The background images for chemiluminescence were obtained by using a shutter in front of the camera. The background measurements for the PLIF diagnostic were done by shifting the laser wavelength to a nonresonant value. Each background image is an average of 200 shots. Additionally, a detector sensitivity correction was performed by dividing all raw images by a flatfield image of the particular camera.

4.1 Instantaneous Flow Field. The results of three typical single-shot PIV measurements at $p=3$ bars, $\Delta p=1\%$, and $\Phi=0.59$ are visualized in Fig. 4. The flow field is depicted in the form of a streamline plot, making the absolute planar velocity of the flow (color coded) and its orientation visible. The first feature that catches one's eye is the inflow region. Its shape changes significantly from one laser pulse to the next. This instationary behavior matches well with the occurrence of vortices distributed statistically in the shear layers. The size of these vortices approximately ranges from 1 cm to 5 cm, and they appear more often in the inner shear layer. The generation of these structures is caused by the different velocities of the injected fuel-air mixture and the hot gas between the inflow regions. The latter moves much slower and its mean flow is directed toward the burner mouth, which in turn is caused by the swirl of the flow and the corresponding conical orientation of the inflow. The backflow rate of reaction products into the burner nozzle is remarkably varying, for example, between Figs. 4(*a*) and 4(*c*). Fewer statements can be done about the outer recirculation zone because only about two-thirds of it are visible due to vignetting by elements of the combustor. The vortices that are involved with the outer recirculation are rarely seen in the PIV single shots. Therefore, it may be presumed that they are generally located very close to the walls of the combustor. Further downstream, in the exhaust gas region on the right-hand side of the analyzed field of view, the planar flow velocity sometimes reaches relatively high values. This flow structure mostly appears with an axially stretched shape. Its radial location as well as its intensity strongly varies. At about half of the single-shot measurements it does not exist at all, whereas other images of the same series come up with a global maximum of velocity in this region. First interpretations of the effect assume this acceleration to be correlated with a fast rotating axial tornadolike vortex core that dominates the transportation of the exhaust gas to the exhaust nozzle of the combustor.

4.2 Instantaneous OH* and OH Distributions. The presence of a high OH* concentration is an indicator of an intensive heat release and therefore usually can be found close to the flame front. The aim of imaging the chemiluminescence of electronically excited radicals is to get an impression of the location and the shape of the zone of the primary reactions. Figure 5 shows a typical instantaneous image of the OH* chemiluminescence at $p=4$ bars, $\Delta p=1\%$, and $\Phi=0.55$. The burner exit is marked by a white ellipse. The inhomogeneous intensity distribution gives an impression of a turbulent flame zone, leading to the assumption of a combustion that is affected by events of local ignition and extinction. Measurements using laser Raman scattering will show whether variations in the mixture fraction might also be a source for the observed inhomogeneities. A good portion of the heat release seems to happen in a region located between 5 cm and 8 cm downstream of the burner exit.

These findings are supplemented by the mappings of the laser-induced OH fluorescence. Detectable concentrations of OH radi-

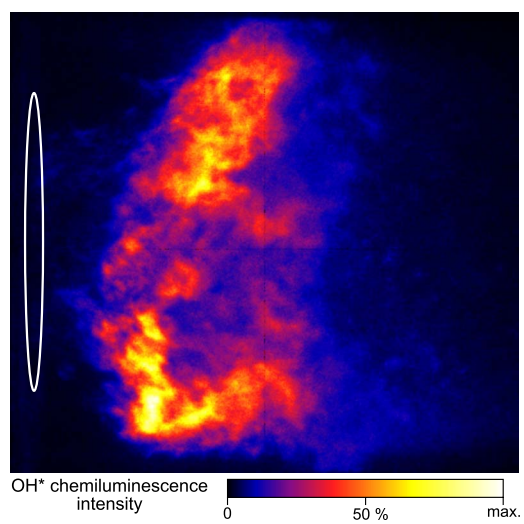

Fig. 5 Instantaneous distribution of the OH* chemiluminescence (line-of-sight) at $p=4$ bars. The white ellipse represents the burner nozzle.

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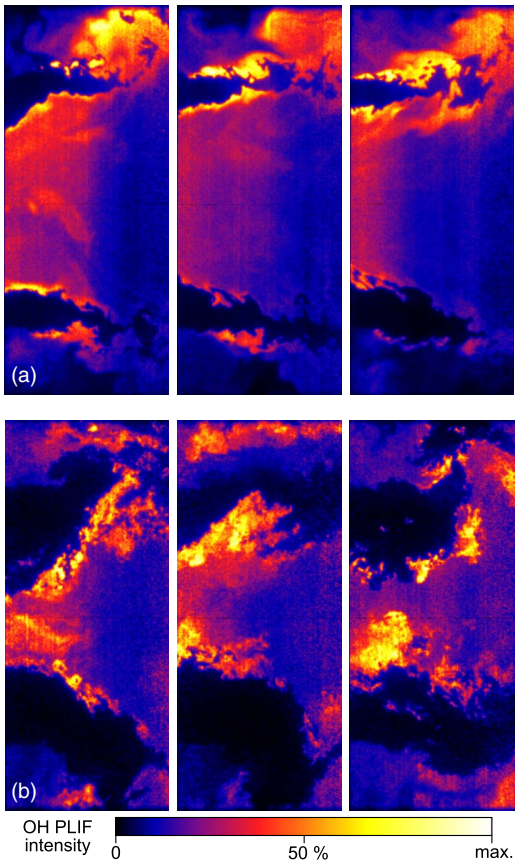

Fig. 6 Single-shot OH PLIF mappings at (*a*) $p=1.5$ bars, $\Delta p=1.1\%$, and $\Phi=0.53$ and (*b*) $p=4$ bars, $\Delta p=3\%$, and $\Phi=0.50$

cals are present not only in the reaction zones but also in the hot exhaust gas when the local temperature is approximately above 1400 K. Since OH is formed in superequilibrium concentrations in the reaction zone, its highest concentrations are found in the flame front. The relaxation toward the equilibrium concentration proceeds via slow three-body reactions and takes a few milliseconds at $p=1$ bar. At a higher pressure it is significantly faster. An extract of a set of single-shot PLIF images is shown in Fig. 6(*a*). The interval between the successive images was about 0.33 s. Only the images from the first laser sheet position are displayed. The pressure drop over the burner was $\Delta p=1.1\%$ at $p=1.5$ bars and $\Phi=0.53$. The black areas near the nozzle exit reflect the region of the inflowing gas of relatively low temperatures. In the inner and outer recirculation zones, medium OH concentrations (blue and red areas, b&w print: dark gray) prevail representing hot gas, probably close to chemical equilibrium. The highest OH concentrations (yellow areas, b&w print: light gray) occur in the shear layers between the inflow and the recirculation zones. These are indicative of the superequilibrium OH and reflect the presence of reaction zones. The "snapshots" clearly show a fragmented flame front.

An example for the influence of the operating parameters on the flame properties can be seen by comparing this set of single shots with the set shown in Fig. 6(*b*). The latter is an extract of a series of PLIF images recorded at $p=4$ bars, $\Delta p=3\%$, and $\Phi=0.50$. The turbulence-related characteristics are much more pronounced as with the operating point from Fig. 6(*a*). The regions that are directly involved in the shear flow are clearly extended, and the superequilibrium OH appears as numerous small spots, each the size of a few pixels. Moreover, excessive OH concentrations occur more frequently in the inner shear layer than in the outer. This indicates a pressure-dependent displacement of the mean distribution of the reaction zones. The distribution of the equilibrium OH in the inner recirculation zone close to the burner changed too. This region has become smaller in diameter. Similar pressure-dependent phenomena were discovered in the temporally averaged OH^* chemiluminescence data, which are treated in Sec. 4.3. A clearer impression of the instationary shape of the reaction zone can be obtained by calculating the gradient of the PLIF intensity. This image processing technique is very sensitive to the rapid growth of the OH concentration between the unburnt mixture and the superequilibrium zone [23]. Such a visualization of a flame at $p=4$, $\Delta p=1\%$, and $\Phi=0.53$ is given in Fig. 7. Here, again, one can see the strongly wrinkled nature of the fragmented flame fronts.

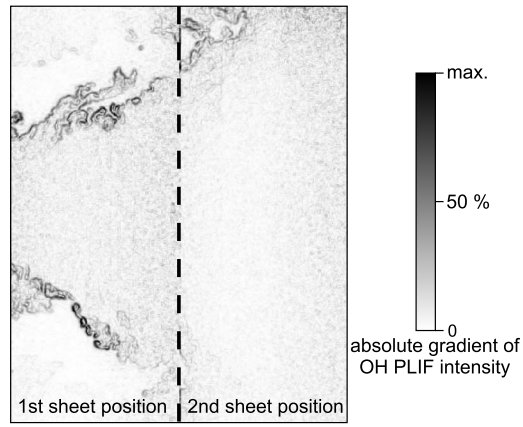

Fig. 7 The absolute gradient of the OH PLIF intensity gives an impression of the instantaneous shape of the flame front. Collage of two measurements with shifted laser sheet positions at different points in time ($p=4$ bars).

4.3 Mean Flow Field. The averaged flow field in Fig. 8 illustrates how the hot exhaust gas in the inner recirculation zone (IRZ) is transported backward until it remixes with the inflow outside or inside of the burner nozzle. A clear distinction between IRZ, outer recirculation zone (ORZ), and inflow region can be obtained by regarding the planar flow direction only (Fig. 9). The sharp contrast in this visualization allows a precise definition of these zones. Remarkably, the complete appearance of the flow field is neither dependent on the pressure inside the combustion chamber nor on the equivalence ratio. The same applies to the variation of the pressure drop over the burner, apart from the

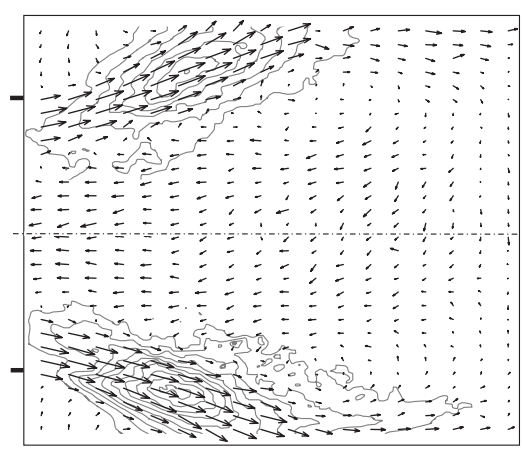

Fig. 8 Mean flow field (sum of correlation) at $p=4$ bars, $\Delta p=1\%$, and $\Phi=0.56$. Background: contour plot of the corresponding mean OH^* density.

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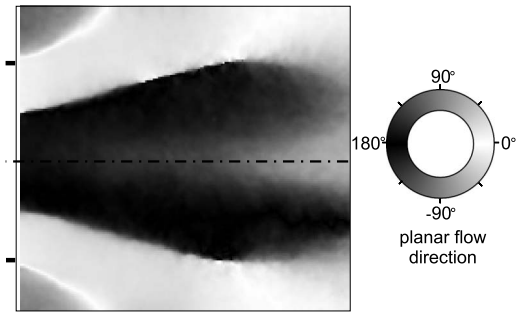

Fig. 9 The inflow region can be clearly distinguished from the recirculation zones by mapping the averaged flow direction

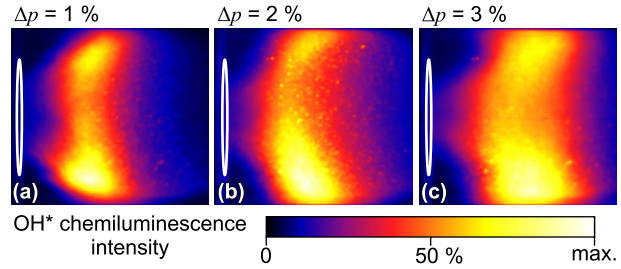

Fig. 12 Line-of-sight mappings of the mean OH* chemiluminescence distribution for different pressure drops over the burner at $p=4$ bars and $\Phi \approx 0.50$. The burner nozzle is marked with a white ellipse.

mass-flow-dependent scaling of the absolute flow speed. In Fig. 10 this scaling is illustrated by plotting the radial profile of the axial flow velocity component averaged over a range of 7–16 mm downstream of the burner.

Explicit findings concerning the turbulence of the flow are obtained by calculating the standard deviation from the mean flow velocity. An appropriate mapping together with the averaged flow vectors is shown in Fig. 11. As expected, strong variations of the velocity can be found within the shear layers. The highest absolute rms values do not coincide exactly with the boundary layer between the IRZ and the inflow standing out in Fig. 9. The maxima are rather located further outside. Another highly turbulent zone is detected further downstream along the central axis of the combustor, although there is no notable shear flow in the temporal average. This can be explained with the existence of a fluctuating axial vortex core, which already was considered at the interpretation of the PIV single-shot results.

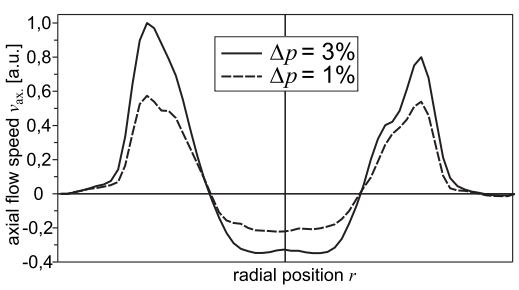

Fig. 10 Radial distribution of the axial component of the flow velocity averaged over an axial range of 7–16 mm downstream of the burner exit

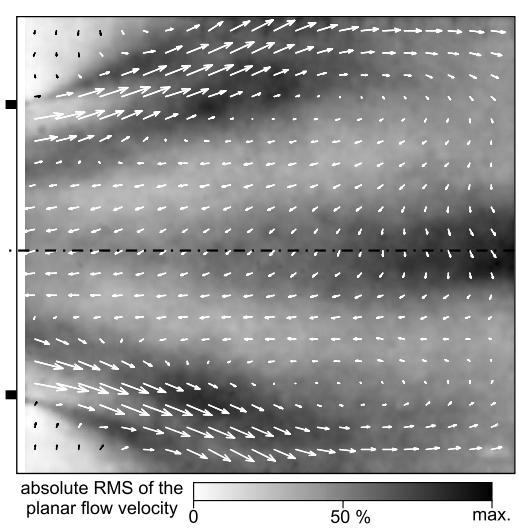

Fig. 11 Mapping of the standard deviation of the flow velocity at $p=3$ bars, $\Delta p=1\%$, and $\Phi=0.59$

4.4 Mean OH* Distribution. In Fig. 12 the temporally averaged chemiluminescence images of three operating points with different mass flows at $p=4$ bars and $\Phi \approx 0.50$ are shown. The asymmetry between the upper and the lower half of the spatial intensity distribution is caused by the inhomogeneous opacity of the optical access. The images reveal the mean flame to stabilize some centimeters downstream of the burner mouth in a ring-shaped region. The position and size of this region are dependent on the mass flow. All of the examined test conditions showed an expansion of the mean flame zone in both the axial and the radial directions when the burner pressure drops, and thus the thermal load is increased. At $\Delta p=3\%$ the region of maximum heat release is located very close to the sidewalls (Fig. 12(*c*)), which may result in a strong thermal load for the combustion chamber. The increase in the flow velocity of the unburnt fuel-air mixture causes a clearly visible downstream shift of the region with maximum heat release and a narrowing of the mean OH* distribution in the IRZ. The latter effect is particularly visible at high pressures.

To be able to quantitatively analyze the spatial information of the chemiluminescence images, the Abel inversion has to be applied first. In order to bypass difficulties with the vignetted flame periphery, it is possible to perform the deconvolution only in an axial range where the radial extension of the flame is small enough for the application of this image processing method. Here, the pressure influence on the radial distribution of the OH* concentration at a distance of 22 mm from the burner was studied. The plots in Fig. 13 were attained by deconvolving the corresponding chemiluminescence images and applying a 9×9 pixel^2 smoothing filter. Additionally, an integration over an axial range of ± 3 mm was done for further noise reduction. Due to the spatial symmetry, this results in two curves of which the arithmetical mean is plotted in the figure. In this case, the window opacity is negligible since the radial extent of the flame is relatively small. The curves of this parameter study (measured at $\Delta p=2\%$, $\Phi=0.53$) clearly reflect a pressure-induced displacement of the OH* concentration from the outer to the inner shear layer. The raw data of the 6 bar measurement had a low signal-to-noise ratio and a considerable inconsistency between the two curves from the opposite sides of the symmetry axis. Therefore, the slight deviation of the 6 bar case from the general tendency must not be overrated. However, a distinct bimodal distribution is only existent at pressures below 4 bars and close to the burner. Further downstream, both peaks merge into a distribution with a single maximum, which coincides with the maximum of the radial distribution of the planar flow velocity. Figure 8 shows a mapping of the mean OH* density in the form of a contour plot together with an overlaid vector field of the mean planar flow velocity. It can be seen from the superimposition that the allocation of the reaction zones is directed by the flow field.

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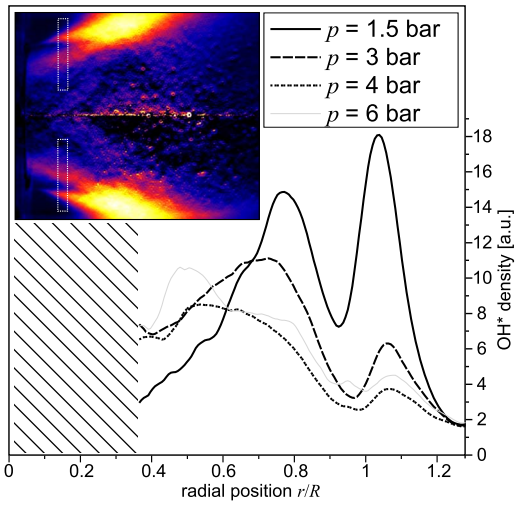

Fig. 13 The radial OH* distribution is influenced by the pressure. The shaded range cannot be analyzed due to deconvolution artifacts. *R* is the burner nozzle radius. Inset: Abel-inverted OH* chemiluminescence. The marked zones were analyzed for the graphs.

5 Summary

An industrial gas turbine burner for lean premixed natural gas flames was equipped with an optical combustion chamber and operated in a high-pressure test rig. The general flame behavior, the structures, and the flow field have been measured using OH* chemiluminescence imaging, planar LIF of OH, and particle image velocimetry. Systematic variations of the mass flow, equivalence ratio, and pressure have been performed. The main goals of this project were a better understanding of the physical and chemical processes dominating the flame behavior and the establishment of an experimental database for the validation of numerical flame simulations.

From the averaged PIV measurement results it was known that the inflow of fresh gases was conically shaped and that the mean flow field had a small ORZ and a large IRZ transporting the hot reaction products back toward the inflow region. Instantaneous PIV measurements exposed the flow in the shear layers to be dominated by small vortices. The PLIF images also showed the unburnt reactants mixing with the combustion products in these regions. By analyzing the distributions of electronically excited OH and superequilibrium concentrations of OH, it could be seen that the ignition happens within the shear layers. Further downstream the primary reactions also took place throughout the complete ring-shaped region with a positive axial flow component. The axial and radial extensions of the averaged flame zone are strongly dependent on the total mass flow. The instantaneous shape of the flame front is disturbed and fragmented by the turbulent flow, especially for conditions with high pressures and high flow velocities. Deconvoluted chemiluminescence images showed a pressure-dependent displacement of the heat release zone from the outer to the inner shear layer. The IRZ of the averaged flow field reaches into the burner nozzle occupying about a quarter of its cross-sectional area. Furthermore, the mean structure of the flow field, e.g., its partition into inflow and recirculation, is independent of the operating conditions. The standard deviation of the flow velocity exposed a relatively strong fluctuating flow within an axially expanded region downstream of the actual flame. This can be interpreted as an effect of a fast rotating exhaust gas swirl.

Altogether, these findings contribute to the understanding of the interaction between the turbulent flow and the chemical reactions in gas turbine combustors.

6 Outlook

In the third measurement phase of this project, laser Raman spectroscopy will be applied to determine the temperature and the concentrations of N_2, O_2, CH_4, H_2, CO_2, CO, and H_2O simultaneously by single-shot measurements with a one-dimensional spatial resolution. For this purpose, a specially designed diagnostic system has already been set up and tested at the German Aerospace Center [24]. Finally, the results of all three measurement phases will be incorporated into a comprehensive database, which will be available for the verification of numerical simulations.

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